



Accelerate Test Execution with Adaptive Test

Reference: RA_453

Unlock Smarter Testing with DAS

Welcome to the Adaptive Test DAS Demo — a powerful illustration of how test programs can be made smarter, more flexible, and responsive to real-time conditions. Whether you're running it locally or envisioning a scaled deployment on remote servers, this demo is built to showcase adaptability.

What This Demo Demonstrates

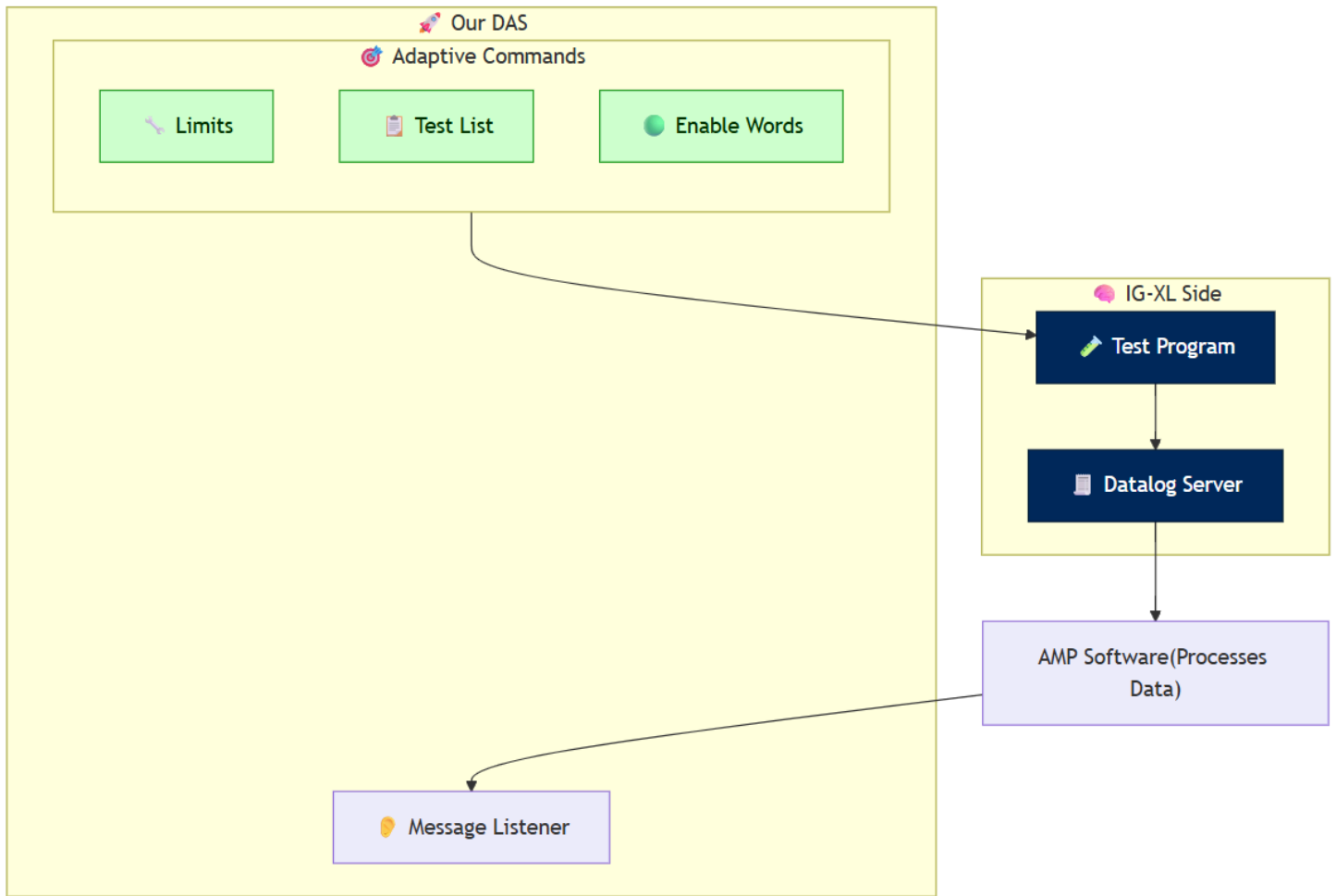
This demo highlights three key capabilities of our Data Analytics Solution (DAS):

1. **Enabling or disabling specific tests** — streamline test execution based on context or priorities
2. **Modifying test limits dynamically** — improve accuracy and adapt to production variability
3. **Controlling enable words** — activate or deactivate sub-flows conditionally

Additionally, the system offers robust observability:

- Monitor messages sent by the tester in real time
- Understand communication patterns driven by adaptive logic

Software Overview

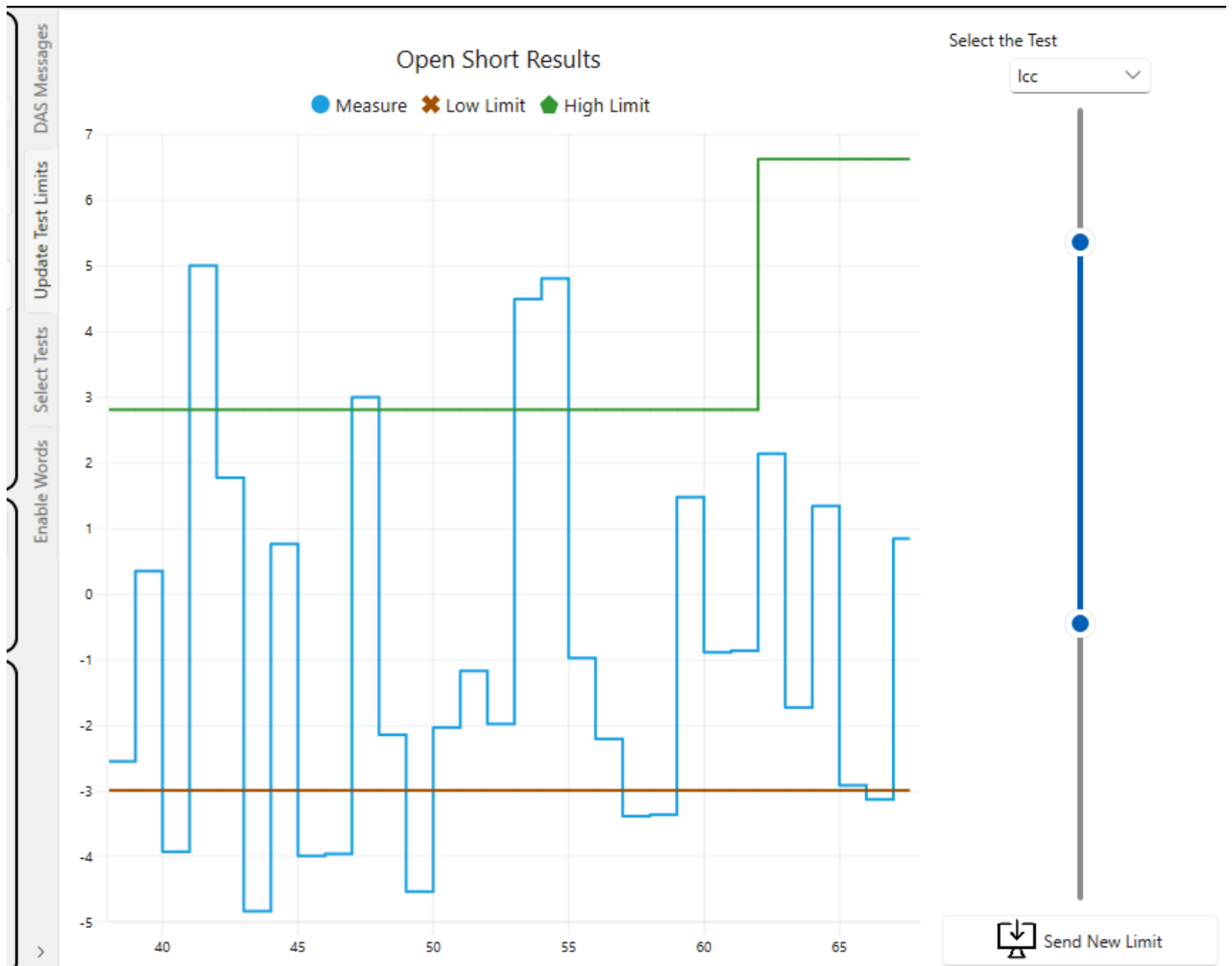


Explore the Demo Software

The demo application includes four dedicated sections:

DAS Section

View all messages generated by the tester. Clicking on a message reveals its detailed contents.



Enable Word Section

The test program includes two enable words that control sub-flow activation. Enable or disable them directly from this interface.

Component	Version
Framework	.NET 8
Environment	Windows 10
Excel	2016 (64-bit)
IGXL	10.60.02

Extend the Demo to Your Environment

To implement a similar system, we recommend reviewing the following guides:

Reference Architecture	Description
RA_0468	How to implement a DAS in C#
RA_0478	How to send an adaptive command in C#

See It in Action

Explore the complete [Instructions to Run the Demo](#) or browse the image gallery to discover key functionalities.

Looking to integrate this approach into your test strategy? [Contact our team](#) to explore how DAS can support your roadmap.

This demo is part of our Reference Architecture Series – built to inspire smarter test development.



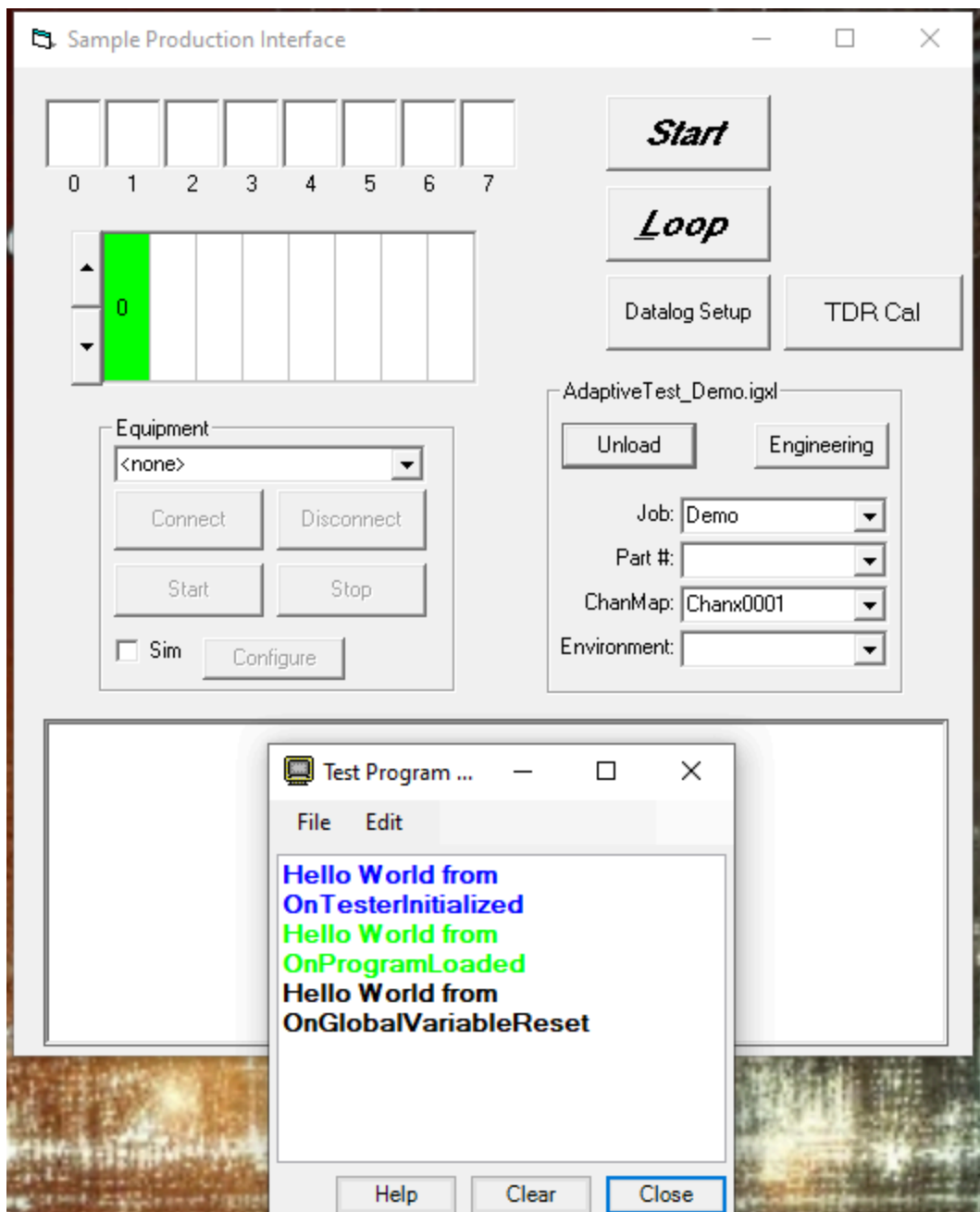
How to Run the Adaptive Test Demo

Reference: RA_453

This section provides step-by-step instructions to explore the key features of the Adaptive Test DAS Demo.

Step 1: Load the Test Program

Use the SimpleOI interface to load your test program.



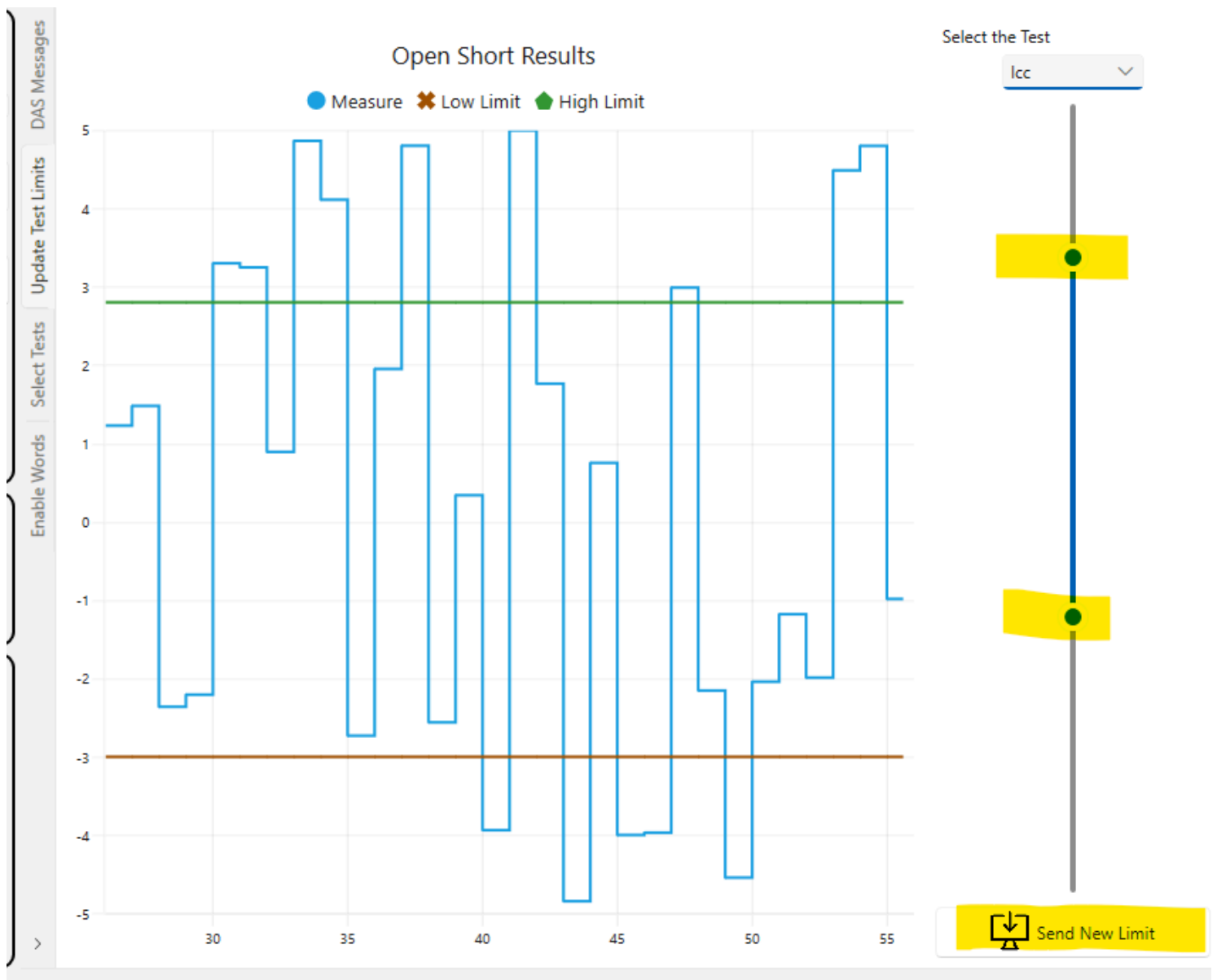
Step 2: Execute the Program in Loop Mode

Click the **Loop** button to execute the program 10 times. Once completed, press the **Stop** button.



Step 3: Adjust Test Limits Dynamically

In the demo application, click **Update Test Limits**. Use the slider to modify the thresholds, then click **Send Limits** to apply them.



Run the test a few times from the SimpleOI interface to observe the impact.

Step 4: Observe the Updated Limits

The graphical output will reflect the new limit values.

DIA

Adaptive Tests

TERADYNE

Communication Status

DAS URL

<http://127.0.0.1:3000/tems/>

Connect the DAS



Disconnect the DAS



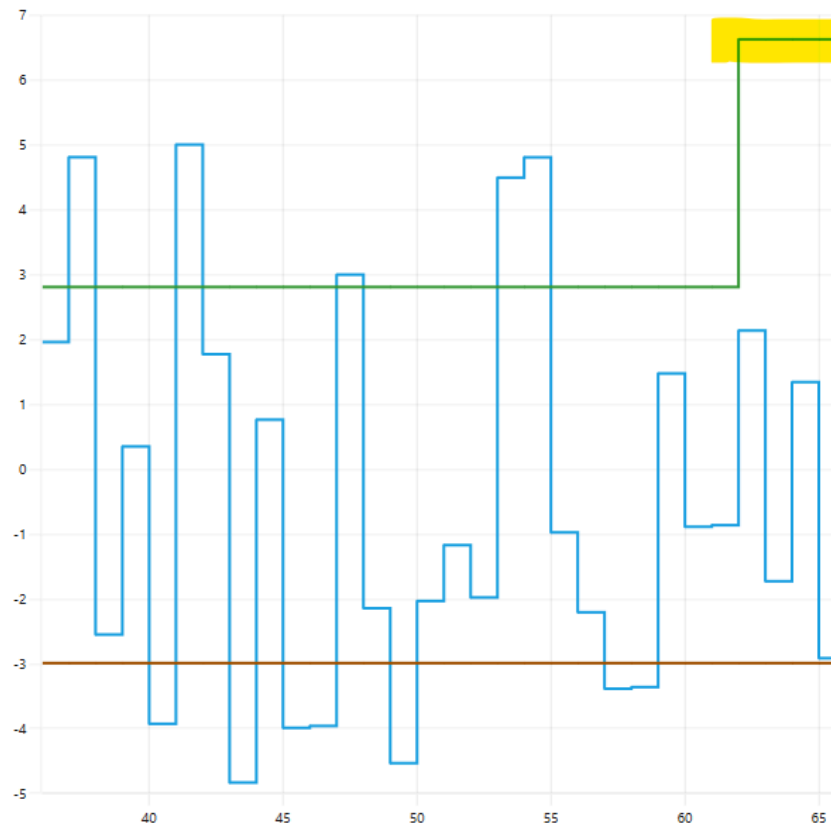
Clear Adaptive Command

Adaptive Command

DAS Messages
Update Test Limits
Select Tests
Enable Words

Open Short Results

● Measure ✕ Low Limit ◆ High Limit



Select the Test

lcc



Send New Limit

In DAS, select the **Adaptive_Change limits** message. Click on the message to inspect its contents.

Time Stamp	Test Cell	Name	Content
08:01:48.9137	TEMS-01	ADAPTIVE_CHANGE	TEP_ACTIONS": [{"ACTION":"UPDATE_LIMITS","REQUEST_ID":1,"TESTSTEP_NAMES": ["Icc"],"LIMITS":{"LIMIT_NAMES":

```
{
  "TIME_STAMP": "2025-05-05T08:01:48.9137009Z",
  "LOT_ID": "UNKNOWNLOT_PID_8380_1",
  "SUBLOT_ID": "UNKNOWNSUBLOT_PID_8380_1",
  "DAS_ID": 1,
  "TESTSTEP_ACTIONS": [
    {
      "ACTION": "UPDATE_LIMITS",
      "REQUEST_ID": 1,
      "TESTSTEP_NAMES": [
        "Icc"
      ],
      "LIMITS": {
        "LIMIT_NAMES": [
          "Icc"
        ]
      }
    }
  ]
}
```

You can also observe the received message in the IGXL test program output.

Teradyne_IG-XL_DataCollect_8380 - OutputWin
File Edit Collect View Help

Site
Sort
Bin

0
1
1

OnProgramEnded: 2025-05-05 10:01:40.390
Adaptive Change: The limits {Icc} on test step {Icc} were changed. New low limit = -3. New high limit = 6.61954453733064.
OnProgramStarted: 2025-05-05 10:01:48.910
Device#: 63

Number	Site	Test Name	Pin	Channel	Low	Measured	High	Force	Loc
<Icc_Fake>									
1000	0	Icc	-1		-3000.0000 mA	2127.3046 mA	6619.5445 mA	0.0000	0

Step 5: Work with Enable Words

This part demonstrates how to activate a sub-flow using enable words. Initially, only the test **ICC** is active.

To explore more:

- Access the engineering view in SimpleOI (password: **Teradyne**)
- Open the test program and locate the **DemoFlow**
- Visualize the flow and locate the enable word logic

AdaptiveTest_Demo.igxl [Book1] - Teradyne IG-XL DataTool - Offline Mode

File Home Insert Page Layout Formulas Data Review View **IG-XL Main** IG-XL Tools Data Analysis Tell me what you want to do

Restart Select Sheet Previous Sheet Apply Style Formulas Insert Delete Run Options Calibrate TDR Validate Job Run Output Window Profiler Off DataCollect Setup Datalog On Job: Demo Channel Map: Chanx00 Part:

DataTool Sheet Run Data Collect

K14 fx

Flow Table							TERADYNE		
							Flow Domain:		
			Gate		Command				
Label	Enable	Job	Part	Env	Opcode	Parameter	TName	TNum	
	OnePin				Test	Icc_Fake		1000	
	DualPin				call	OnePin_Flow			
					call	DualPin_Flow			
					set-device				
					stop				

Click on the **OnePin EnableWord** option and run the test program once.

Select the enable words you want to enable or disable

☒ Enable or Disable the enable word : OnePin

☐ Enable or Disable the enable word : DualPin

You'll see that more tests are now activated:

Adaptive Change: The following enable words were enabled: {OnePin}.

OnProgramStarted: 2025-05-05 10:09:38.800
Device#: 68

Number	Site	Test Name	Pin	Channel	Low	Measured	High	Force	Loc
<Icc_Fake>									
1000	0	Icc	-1		-3000.0000 mA	833.5900 mA	6619.5445 mA	0.0000	0
OnePin									
<OnePin_NoHighLimit>									
200	0	OnePin_NoHighLi	-1		100.0000 m	1.1235	N/A	0.0000	0
<OnePin_NoLowLimit>									
300	0	OnePin_NoLowLim	-1		N/A	500.0000 m	900.0000 m	0.0000	0
<OnePin_NoLimit>									
400	0	OnePin_NoLimit	-1		N/A	-700.0000 m	N/A	0.0000	0
<OnePin_MilliAmpUnit>									
500	0	OnePin_MilliAmp	-1		-1000000000.0000 nA	17500000000.0000 nA	(F) 0.0000 nA	0.0000	0

A new Adaptive Change message will also be generated:

08:09:38.8061	TEMS-01	ADAPTIVE_CHANGE	ID_8380_1","SUBLOT_ID":"UNKNOWNSUBLOT_PID_8380_1","DAS_ID":1,"TESTSTEP_ACTIONS":[],"ENABLEWORD_ACTIONS":[{"ACTION":"ENABLE","REQUEST_ID":2,"ENABLEWORDS":
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```

"TIME_STAMP": "2025-05-05T08:09:38.806154Z",
"LOT_ID": "UNKNOWNLOT_PID_8380_1",
"SUBLOT_ID": "UNKNOWNSUBLOT_PID_8380_1",
"DAS_ID": 1,
"TESTSTEP_ACTIONS": [],
"ENABLEWORD_ACTIONS": [
  {
    "ACTION": "ENABLE",
    "REQUEST_ID": 2,
    "ENABLEWORDS": [
      "OnePin"
    ],
    "STATUS": "OK"
  }
]

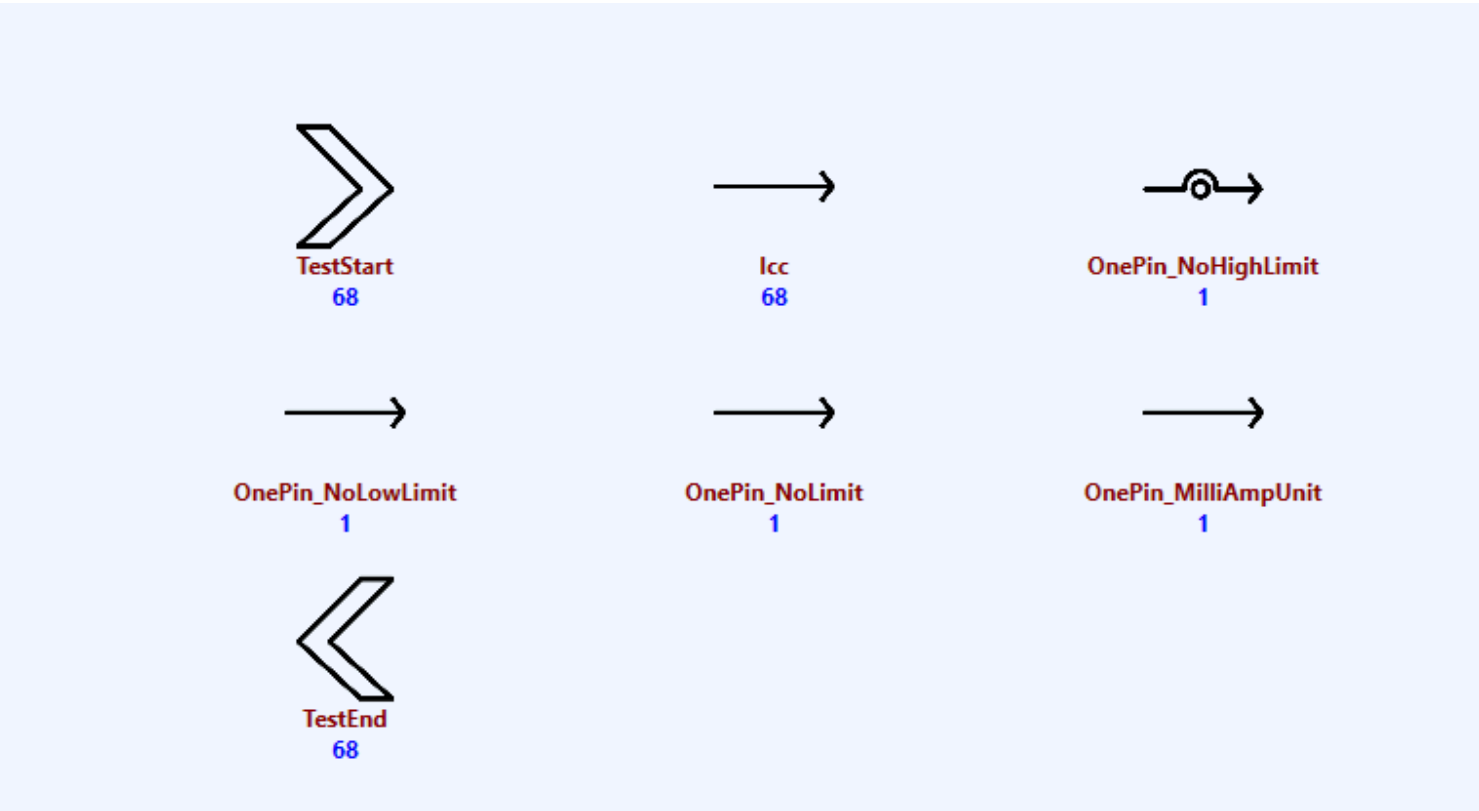
```

Step 6: Enable or Disable Tests

This step shows how individual tests can be selectively enabled or disabled.

- Go to the **Select Test** tab in the DAS interface
- A count of executions for each test is shown in blue
- For best demo results, activate **DoAll** in IGXL Run Options

Now, disable the `OnePin_NoHighLimit` test by selecting it.



Run the test program once. You'll see that the test is skipped.

08:25:20.0344	TEMS-01	ADAPTIVE_CHANGE	ID_8380_1";SUBLOT_ID":"UNKNOWN";SUBLOT_PID_8380_1";DAS_ID":1,"TESTSTEP_ACTIONS":[{"ACTION":"DISABLE","REQUEST_ID":3,"TESTSTEP_NAMES":
<pre>{ "TIME_STAMP": "2025-05-05T08:25:20.034446Z", "LOT_ID": "UNKNOWNLOT_PID_8380_1", "SUBLOT_ID": "UNKNOWN";SUBLOT_PID_8380_1", "DAS_ID": 1, "TESTSTEP_ACTIONS": [{ "ACTION": "DISABLE", "REQUEST_ID": 3, "TESTSTEP_NAMES": ["OnePin_NoHighLimit"] }], "STATUS": "OK" }</pre>			

You can confirm this in IGXL:

OnProgramEnded: 2025-05-05 10:09:38.870
Adaptive Change: The following test steps were disabled: {OnePin_NoHighLimit}.
OnProgramStarted: 2025-05-05 10:25:20.030
Device#: 69

Number	Site	Test Name	Pin	Channel	Low	Measured	High	Force	Loc
<Icc_Fake>									
1000	0	Icc	-1		-3000.0000 mA	-4192.8535 mA (F)	6619.5445 mA	0.0000	0
OnePin									
<OnePin_NoLowLimit>									
300	0	OnePin_NoLowLim	-1		N/A	500.0000 m	900.0000 m	0.0000	0
<OnePin_NoLimit>									
400	0	OnePin_NoLimit	-1		N/A	-700.0000 m	N/A	0.0000	0
<OnePin_MilliAmpUnit>									
500	0	OnePin_MilliAmp	-1		-1000000000.0000 nA	17500000000.0000 nA (F)	0.0000 nA	0.0000	0

Download the Demo

[Download the demo from here](#)